MATHEMATICS

Math's Guide: Reasoning & Problem Solving

Welcome to your Maths Reasoning Handbook!

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1. The Maths Detective Toolkit (TWM Framework)

In Mathematics, being a 'Maths Detective' means showing HOW and WHY an answer works. The Cambridge curriculum measures four key Thinking and Working Mathematically (TWM) skills:

TWM Skill	What it Means (Plain English)	Detective Action
Conjecturing	Making an educated guess or mathematical prediction based on clues.	Test your prediction with numbers!
Generalising	Spotting a pattern or rule that works every single time.	Ask: Does this rule always work?
Convincing	Explaining and proving your solution step-by-step using math words.	Write: 'I know this because...'
Characterising	Identifying the special properties or features of a mathematical object.	Compare numbers and spot differences.

💬 Sentence Starters for Reasoning Questions

Use these starting phrases to write clear, full-mark mathematical explanations:

- 'My estimate is _____ because _____.'
- 'The answer cannot be _____ because _____.'
- 'I know these two values are equal because _____.'
- 'The mean changed from _____ to _____ because adding the value _____.'
- 'I can prove this by converting _____ into a percentage (_____%).'

2. Addition & Subtraction Reasoning

Addition and subtraction questions go beyond simple calculations. You will solve missing-digit puzzles, spot calculation errors, calculate across zero with negative numbers, and compute multi-step decimal problems up to 3 decimal places.

Word Bank: Addition & Subtraction

- Sum: The total when two or more numbers are added together.
- Difference: The amount left when one number is subtracted from another.
- Inverse Operation: The opposite operation (Addition <--> Subtraction) used to check answers.
- Decimal Place: The position of a digit to the right of the decimal point (e.g. 0.125 has 3 decimal places).
- Negative Number: A number less than zero (e.g. -5°C).

Missing Digit Column Calculations

Problem: Fill in the missing digits [?] in this column addition:

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  4.[?]82
+3.5 6[?]
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  8.2 4 7
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Strategy & Explanation:

1. Look at the thousandths column: 2 + [?] = 7. Therefore, the missing digit is 5.
2. Look at the hundredths column: [?] + 6 = 4. Since 6 is larger than 4, it must regroup to 14. [?] + 6 = 14, so missing digit = 8 (with 1 carried over).
3. Check tenths column: 1 (carried) + 8 + 5 = 14 (write 4, carry 1).
4. Check units column: 1 (carried) + 4 + 3 = 8. Perfect!

Temperature Jumps & Negative Numbers

Problem: At midnight in Manchester, the temperature was -8°C. By midday, the temperature rose by 15°C. What was the midday temperature?

Strategy: Break the jump into two steps using zero as a bridge:
- Step 1: From -8°C to 0°C is a rise of +8°C.
- Step 2: Remaining rise = 15°C - 8°C = +7°C.
- Midday Temperature = +7°C.

⚠️ Spot the Mistake Challenge! (Column Subtraction)

A student solved 12.45 - 3.829 and got 9.631. What mistake did they make?

- The Mistake: The student forgot to place a placeholder zero on 12.450, and simply brought down the 9 without regrouping (9 - 0 = 9).
- Correct Solution: 12.450 - 3.829 = 8.621.

3. Averages & Data Mastery (Mean, Mode, Median, Range)

In Stage 6, you must know how to calculate the mean, median, mode and range, as well as choosing which average is best for a given set of data.

Choosing the Best Average

Average / Measure	How to Calculate	Best Used When...
Mean	Add all numbers, then divide by the total count.	Finding an overall equal share across consistent data.
Median	Order numbers from smallest to largest; find the middle value.	Data contains extreme outliers (very high or low numbers).
Mode	Find the value that appears most frequently.	Data is discrete or categorical (e.g., shoe sizes, colors). (This information is from Unit 15 – Data)

Working Backwards from a Known Mean

Problem: The mean score of 5 test papers is 18. Four of the scores are 15, 20, 17, and 19. What is the 5th score?

Detective Working Steps:
1. Find the required total sum: Total Sum = Mean x Count = 18 x 5 = 90.
2. Add the 4 known scores: 15 + 20 + 17 + 19 = 71.
3. Subtract to find the missing score: 90 - 71 = 19.
• The 5th score is 19.

4. Fractions, Decimals & Percentages

Understanding equivalence between fractions, decimals, and percentages allows you to compare and order values quickly.

Fraction	Decimal	Percentage
1/2	0.5	50%
1/4	0.25	25%
3/4	0.75	75%
1/5	0.2	20%
1/10	0.1	10%
1/100	0.01	1%

Ordering Mixed FPD Values

Problem: Order these values from smallest to largest: 3/5, 0.62, 65%.

Detective Strategy: Convert all values into percentages:
• 3/5 = 6/10 = 60%
• 0.62 = 62%
• 65% = 65%
Ordered from smallest to largest: 3/5 (60%), 0.62 (62%), 65%.

5. End-of-Guide Practice Challenge

Test your skills! Complete these 9 reasoning questions. Write down your full step-by-step mathematical justifications.

Addition & Subtraction

Q1. A mountain climber starts at -150 meters elevation in a valley and climbs up 425 meters. What is their final elevation?

Q2. Fill in the missing digits: 5 [?] . 4 8 + 2 . 7 [?] = 8 . 2 1

Q3. Sam buys items costing $4.85, $12.350, and $0.95. He pays with a $20 note. How much change does he receive?

Averages & Data

Q4. The mean height of 4 children is 135 cm. A 5th child who is 150 cm tall joins the group. What is the new mean height?

Q5. A shoe shop sells sizes: 4, 5, 5, 5, 6, 6, 7, 11. Which average (mean, median, or mode) should the owner use to stock the most popular size? Explain why.

Q6. Five numbers have a median of 12 and a mode of 8. The smallest number is 8. Write down a possible set of 5 numbers.

Fractions, Decimals & Percentages

Q7. Put these in order from largest to smallest: 7/10, 0.72, 68%, 3/4.

Q8. A jacket costs $80. It is on sale for 25% off. How much does the jacket cost after the discount?

Q9. In a class of 30 pupils, 2/5 walk to school and 30% take the bus. How many pupils travel by other means?

Solution Guide & Full Answer Key

Solutions to Section A: Addition & Subtraction

• A1 Answer: +275 meters. (Working: -150 + 425 = 425 - 150 = 275m).

• A2 Answer: 5.48 + 2.73 = 8.21. Missing digits are 4 and 3.

• A3 Answer: $1.85 change. (Total cost: 4.85 + 12.35 + 0.95 = $18.15. Change: 20.00 - 18.15 = $1.85).

Solutions to Section B: Averages & Data

• B4 Answer: 138 cm. (Original total sum = 135 x 4 = 540 cm. New total = 540 + 150 = 690 cm. New mean = 690 / 5 = 138 cm).

• B5 Answer: Mode. (Reasoning: Mode represents the most frequently bought size (size 5). The mean would be distorted by size 11).

• B6 Answer: Example set: 8, 8, 12, 14, 15. (Must have two 8s, middle number 12, and two numbers > 12).

Solutions to Section C: Fractions, Decimals & Percentages

• C7 Answer: 3/4 (75%), 0.72 (72%), 7/10 (70%), 68%.

• C8 Answer: $60. (Discount = 25% of $80 = $20. Sale price = 80 - 20 = $60).

• C9 Answer: 9 pupils. (Walkers = 2/5 of 30 = 12. Bus takers = 30% of 30 = 9. Total = 21. Other = 30 - 21 = 9 pupils).